

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book is to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following:

A Answer the Following:

- I) Discuss the relationship between biological action and physicochemical properties for ANY ONE of the following with suitable example: **05**
1. Partition Coefficient
 2. Hydrogen bonding

II) Explain any two Phase II metabolic reactions along with suitable examples. **05**

B Write classification, SAR and metabolism of cholinergic drug. Give the detail scheme for synthesis of Pyridostigmine. 10

C Write classification, SAR and chemical mechanism of action of sympathomimetic. 10
Give detail scheme for synthesis of Isoprenalolol.

Q 2 Attempt any TWO of the following:

A Write classification, SAR and chemical mechanism of action of Antiasthmatic agents. 10
Give synthesis of Salbutamol.

B Write a Short Note on:

- I) Antiemetic agents **05**
- II) Respiratory Stimulants **05**

C Answer the Following:

- I) Antispasmodic agents **05**
- II) Pro-drugs **05**

SECTION- II

Q 3 Attempt any TWO of the following:

A Write the comments on :(ANY THREE)

10

1. Pyridine is more basic than pyrrole.
2. Electrophilic substitution reaction in furan occurs at 2nd and 5th position.
3. Impurities of thiophene in benzene can be removed by cold sulphuric acid.
4. Imidazole is more basic than pyrazole.

B Answer the Following:

I) Give synthesis of Indole formation along with its reaction mechanism.

05

II) Give the structure of:

05

(1) Piperidine (2) Pyrazine (3) Isoquinoline (4) Triazole, (5) Pyran

C Answer the Following:

I) Write Skraup synthesis with reaction mechanism.

05

II) Write Paal-Knorr synthesis for Pyrrole with reaction mechanism.

05

Q 4 Attempt any TWO of the following:

A I) Write the comments on :(ANY TWO)

05

1. Proton pump inhibitors act as pro-drug.
2. Thiazole is less basic than pyridine.
3. Thiophene is more aromatic than Furan and Pyrrole.

II) Write a brief note on Antitussives Agents.

05

B Write classification, SAR and chemical mechanism of action of Proton Pump Inhibitors. Give scheme of synthesis of Pantoprazole.

10

C Answer the Following:

I) Give any TWO methods of preparation of the following:

05

(1) Oxazole (2) Furan (3) Thiophene (4) Indole

II) Give Proton Transfer (Hantzsch) synthesis for Pyridine with reaction mechanism.

05